

Netflix User Data Dashboard



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Background

What is the problem?

- Netflix and streaming platforms rely on understanding user behavior to retain subscribers

Why is it interesting?

- High churn rates directly impact revenue and customer engagement.

Goal: Analyze watch history, movies, and user information to uncover engagement patterns through user segmentation and churn prediction model

Data & Method

Dataset: 210K records of Netflix Users [Dataset Link](#)

Data Preparation:

- Handle missing values and inconsistencies
- Encode categorical features
- Feature engineering
- Aggregate watch history and movie per user for modeling

Model:

- User segmentation: K-Means clustering to identify behavior patterns
- Churn prediction: Logistic regression, Random Forest, and XGBoost

Evaluation Metrics: Recall, F1-score

Limitation:

- Synthesized data

Feature Engineering

Biggest challenge was understanding what metrics to use for our model, we needed a business mindset.

- Improved metrics to measure customer behavior
 - User recency
 - Recent engagement in the last 7 days or 30 days (rolling window)
 - Lifetime watch metrics
- User segmentation (RFE segmentation)
 - Pick between geographic, **behavior**, or personal data
 - Recency -> `days_since_last_watch`
 - Frequency -> `total_sessions`
 - Engagement -> `avg_completion_rate`

Use Cases

1. Marketing Manager (Retention Campaigns)

- **Objective:** Identify high-risk subscribers.
- **Action:** Filters users by churn probability > 85% and export for targeted offers.
- **Outcome:** Supports marketing strategies to retain subscribers.

2. Studio Executive (Content Investment)

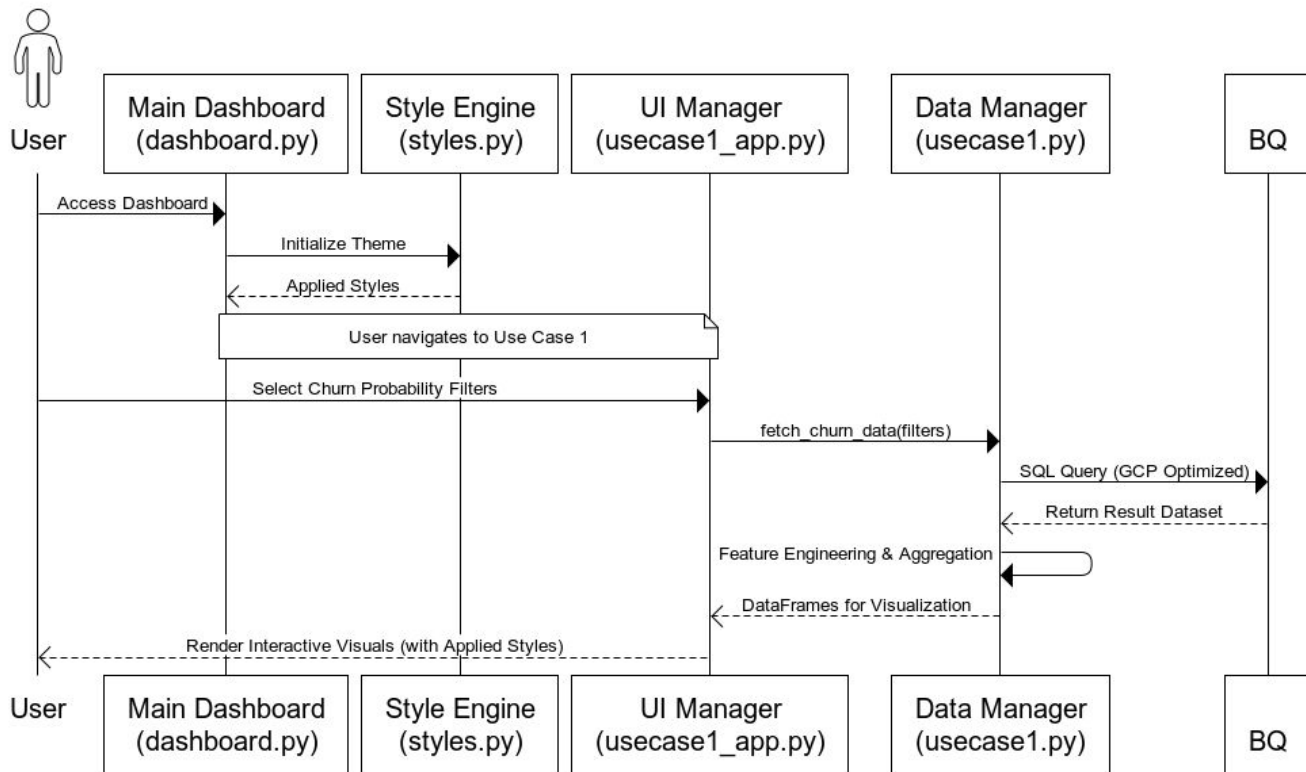
- **Objective:** Evaluate content performance.
- **Action:** Analyze retention impact of high-rated content vs content volume.
- **Outcome:** Informs decisions on content renewal and budget allocation.

3. Product Manager (Cause of Churn)

- **Objective:** Identify underperforming platform features.
- **Action:** Analyze search logs and recommendation drop-offs.
- **Outcome:** Improves user experience by optimizing feature performance.

Design

Use Case Interaction: Churn Probability Analysis

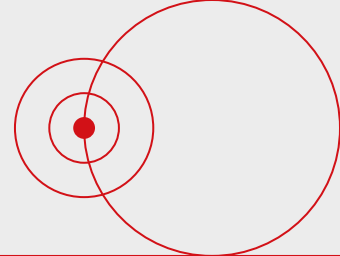


Demo

Lessons learned & Future Work

- Importance of Modular Encapsulation
 - By isolating each use case into independent directories, we minimized code dependency. This High Cohesion allowed team members to develop features simultaneously without affecting the core system stability.
- Effective Separation of Concerns
 - Decoupling the data fetching logic (GCP BigQuery) from the presentation layer (Streamlit) was vital. This design made it much easier to debug database connection issues without interfering with the UI code.
- Configuration over Hard-coding
 - Managing environment variables (GCP Project IDs) taught us that hard-coding sensitive information is a major technical debt. Moving toward .env files improved the Portability and security of our software.
- Automated Configuration Scripting
 - Developing a CLI tool or script that automatically configures the environment variables (like GCP PROJECT_ID) based on the user's local setup to minimize manual configuration errors.
- Improving Test Coverage (Integration Tests)
 - Expanding tests to ensure data integrity and edge-cases handling for all BigQuery responses.
- Logging & Monitoring Dashboard
 - Integrating a centralized logging system to monitor application health and track query performance in real-time.

Key Takeaways and Next Steps



Challenge 1

List any potential challenges and obstacles going into next quarter. Identify potential roadblocks to help inform planning. Keep it short – you can expand in the following slides.

Link to resource

Challenge 2

Describe another potential challenge going into next quarter.

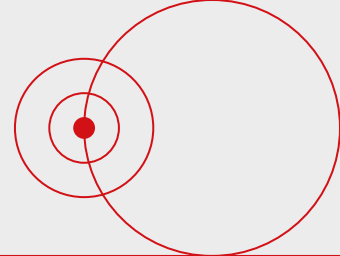
Link to resource

Challenge 3

Describe another potential challenge going into next quarter.

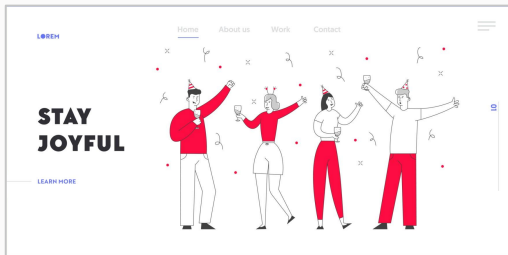
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Challenge 1



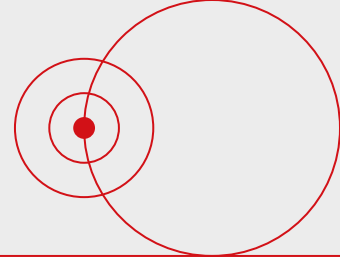
Dive deeper into a challenge. Explain why this challenge is important and how it may impact the upcoming quarter.

To help your audience think about the challenge and what it means for your business, use this space to provide more context, suggest root causes, add concrete data or key performance indicators, etc.



Link to resource

Challenge 2



Describe another challenge. For example, this could be an operational issue. Discuss what led to the issue, and the steps you'll take to avoid it in the next quarter.

Step 1

Outline what needs to be done to prevent this issue from recurring.

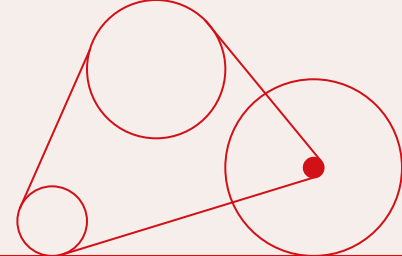
Step 2

Specify the teams responsible for each step in the action plan.

Step 3

Give a timeline in which these tasks must be completed.

Goal 1



Expand on one of your goals. In a few words, explain why this goal is important and how it'll be measured. Include a high-level overview of the steps you'll take to achieve this goal.

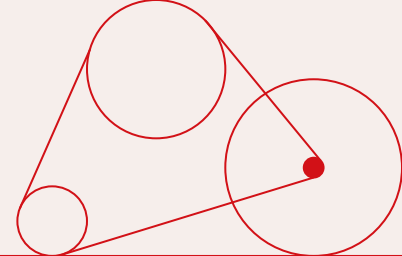
KPI 1

Illustrate your goal with a specific metric or statistic.

KPI 2

Illustrate your goal with a specific metric or statistic.

Goal 2



Briefly describe another goal and explain how it will contribute to your overall objectives for the quarter.

1

Outline the timeline for the goal, including the major milestones and deadlines.

2

Explore the KPIs that will be used to measure progress towards the goal.

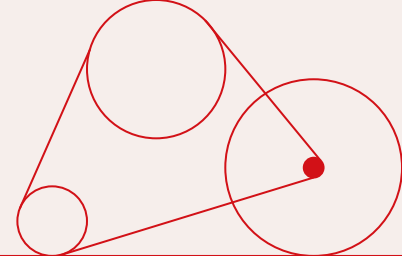
3

Mention the teams involved.

4

Discuss any potential roadblocks to reaching the goal and how you hope to avoid them.

Goal 3



Explore another goal for the quarter, including why it's a priority, how you intend to achieve it, and the targets you hope to reach.



Upcoming project 1

Break down an upcoming project, including:

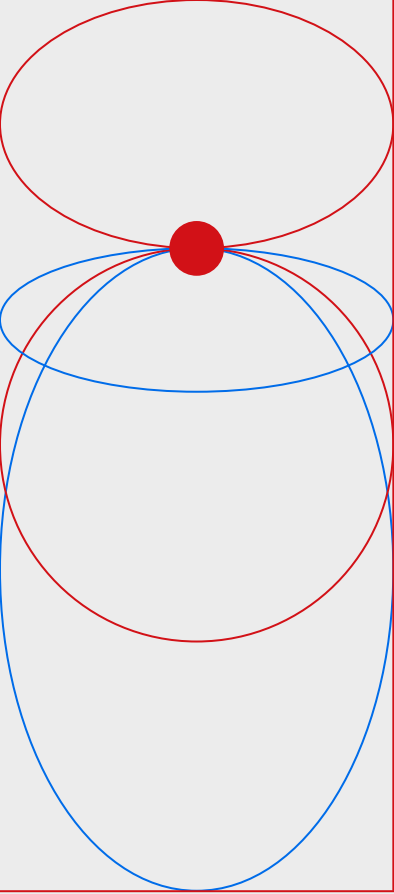
- The expectations for the project
- The teams involved
- The metrics you'll measure



Upcoming project 2

Break down an upcoming project, including:

- The expectations for the project
- The teams involved
- The metrics you'll measure



*Thank
you*